Name - Smruthi Bangalore Thandava Murthy

<u>Result Snapshots</u>

I was able to extract all the tables in the 3 pages of the original pdf into csv files

Input Image 1:

Table 1
Groundwater Analytical Results
RACER - Flint West # 12990

			Sample ID	MW-100S	MW-101S	MW-102S	MW-103S	MW-104S	MW-105S	MW-106SR	MW-107S	MW-108S	MW-109	MW-110S	MW-111S	MW-112S	MW-113S	MW-114S	Dupe 1	Dupe 2	Field Blank	Equip Blank	Field Blank	Equipment Blank	Trip	Equip	Field
			Date Collected	02/25/2020	02/25/2020	02/25/2020	02/28/2020	02/25/2020	02/25/2020	02/25/2020	02/25/2020	02/27/2020	02/27/2020	02/27/2020	02/27/2020	02/27/2020	02/27/2020	02/27/2020	02/27/2020	02/27/2020	02/25/2020	02/25/2020	02/27/2020	02/27/2020	02/28/2020	02/28/2020	02/28/2020
METALS ANALYTE (ug/L)	DW	GSI																	MW-112S	MW-114S							
Arsenic, Dissolved	10	10		<0.002	<0.002	<0.002	4	<0.002	<0.002	<0.002	<0.002	0.257	9	0.518	<0.002	13	1.511	25	13	24	NA	NA	NA	NA	NA	NA	NA
Arsenic	10	10		<0.002	<0.002	<0.002	46	<0.002	0.316	1.173	0.465	<0.002	69	2	0.998	41	3	62	43	69	NA	NA	NA	NA	NA	NA	NA
Chromium, Dissolved	100	160	G	0.275	0.891	7	0.204	0.562	0.463	0.719	<0.005	0.295	0.138	0.312	6	0.214	0.518	13	0.39	<0.005	NA	NA	NA	NA	NA	NA	NA
Chromium	100	160	G	0.42	2.753	6	0.388	16	280	764	0.185	0.431	1.767	0.862	6	0.349	6	0.97	0.356	1.084	NA	NA	NA	NA	NA	NA	NA
Chromium VI, Dissolved	100	160		<0.005	<0.005	8	<0.01	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	4	<0.005	<0.005	<0.005	<0.005	NA	NA	NA	NA	NA	NA	NA
Chromium VI	100	160		<0.01	<0.01	<0.01	<0.02	<0.01	<0.01	<0.01	4	<0.025	<0.5	<0.5	<0.5	<0.5	<0.025	<0.5	<0.5	<0.5	NA	NA	NA	NA	NA	NA	NA
Copper, Dissolved	1000	20	G	0.405	<0.005	<0.005	<0.005	3.473	0.396	0.576	0.86	1.357	<0.005	0.835	0.501	<0.005	0.492	0.751	0.391	0.611	NA	NA	NA	NA	NA	NA	NA
Copper	1000	20	G	<0.005	0.698	0.516	0.528	9	6	14	0.996	0.564	<0.005	0.835	0.694	<0.005	0.973	6	<0.005	6	NA	NA	NA	NA	NA	NA	NA
Lead, Dissolved	4	44	G	<0.003	<0.003	<0.003	0.208	<0.003	<0.003	<0.003	<0.003	<0.003	<0.003	<0.003	<0.003	<0.003	<0.003	<0.003	0.584	<0.003	NA	NA	NA	NA	NA	NA	NA
Lead	4	44	G	0.249	0.283	<0.003	<0.003	<0.003	<0.003	0.886	0.283	<0.003	<0.003	0.242	0.339	<0.003	0.4	4	<0.003	4	NA	NA	NA	NA	NA	NA	NA
Selenium, Dissolved	50	5		<0.005	<0.005	<0.005	<0.005	<0.005	9	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	NA	NA	NA	NA	NA	NA	NA
Selenium	50	5		<0.005	<0.005	<0.005	<0.005	<0.005	9	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	<0.005	NA	NA	NA	NA	NA	NA	NA
Zinc, Dissolved	2400	260	G	6	4.654	5	2.299	9	1.504	1.216	9	7	3.151	3.037	6	2.665	8	27	6	29	NA	NA	NA	NA	NA	NA	NA
Zinc	2400	260	G	1.56	2.074	1.632	2.05	14	1.948	7	6	2.742	2.004	1.879	1.594	1.588	6	44	1.221	43	NA	NA	NA	NA	NA	NA	NA
VOC ANALYTE (ug/L)	DW	GSI																	MW-112S	MW-114S							
Diethyl ether	10 (E)	ID		<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
Acetone	730	1,700		<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50
Methyl iodide	NC	NC		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Carbon disulfide	800	NC		<5	<5	<5	0.26	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	0.15	0.18	0.17	0.17
tert-Methyl butyl ether (MTBE)	40 (E)	7,100 (X)		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Acrylonitrile	2.6	2.0 (M); 1.2		<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2
2-Butanone (MEK)	13,000	2,200		<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25	<25
Dichlorodifluoromethane	1,700	ID		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Chloromethane	260	NC		0.15	<5	<5	<5	<5	<5	<5	<5	<5	0.13	<5	<5	<5	0.44	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Vinyl chloride	2	13		<1	<1	<1	0.81	<1	<1	<1	<1	<1	18	<1	<1	10	0.79	5	11	3	<1	<1	<1	<1	<1	<1	<1
Bromomethane	10	4.2; [5(M)]		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Chloroethane	430	1,100		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	0.39	<5	<5
Trichlorofluoromethane	2,600	NA		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
1,1-Dichloroethene	7	130		<1	<1	<1	<1	<1	<1	<1	<1	<1	1	<1	<1	1	<1	0.84	1	0.82	<1	<1	<1	<1	<1	<1	<1
Methylene chloride	5	1,500		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	0.11	<5	<5	0.13	<5	0.43	0.41	0.36	0.76	0.22	0.36	0.45
trans-1,2-Dichloroethene	100	1,500		<1	<1	<1	<1	<1	<1	<1	<1	<1	0.5	<1	<1	0.28	<1	0.78	0.24	0.83	<1	<1	<1	<1	<1	<1	<1
1,1-Dichloroethane	880	740		<1	<1	<1	<1	<1	<1	<1	<1	<1	1	<1	<1	0.99	<1	3	1	2	<1	<1	<1	<1	<1	<1	<1
cis-1,2-Dichloroethene	70	620		9	<1	<1	<1	<1	<1	<1	0.11	<1	17	<1	0.35	3	0.44	127	3	125	<1	<1	<1	<1	<1	<1	<1
Tetrahydrofuran	95	11,000		<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	<90	0.22	<90	<90	<90	<90	<90
Chloroform	80	350		<1	<1	<1	<1	0.71	<1	0.24	<1	0.81	<1	<1	<1	<1	<1	0.22	<1	0.23	<1	<1	<1	<1	<1	<1	<1
Bromochloromethane	NC	NC		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
1,1,1-Trichloroethane	200	89		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	2	<1	2	<1	<1	<1	<1	<1	<1	<1
4-Methyl-2-pentanone (MIBK)	1800	ID		<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50
2-Hexanone	1000	ID		<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50
Carbon tetrachloride	5.0	38	X	<1	<1	<1	<1	<1	<1	<1	<1	2	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Benzene	5.0	200		<1	<1	<1	0.26	<1	<1	<1	<1	<1	<1	<1	<1	0.2	<1	<1	0.22	<1	<1	<1	<1	<1	<1	<1	<1
1,2-Dichloroethane	5.0 (A)	360 (X)		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Trichloroethene	5.0	200		16	2	0.16	<1	<1	<1	<1	0.15	0.14	13	<1	6	6	7	335	6	350	<1	<1	<1	<1	<1	<1	<1
1,2-Dichloropropane	5.0 (A)	230 (X)		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Bromodichloromethane	80.0	NC		<1	<1	<1	<1	0.17	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Dibromomethane	230	NA		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
cis-1,3-Dichloropropene	NC	NC		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Toluene	790	270		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
trans-1,3-Dichloropropene	NC	NC		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
1,1,2-Trichloroethane	5.0 (A)	330 (X)		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Tetrachloroethene	5.0	60		<1	<1	<1	<1	<1	74	<1	<1	<1	<1	<1	<1	<1	<1	0.2	<1	0.18	<1	<1	<1	<1	<1	<1	<1
trans-1,4-Dichloro-2-butene	NC	NC		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Dibromochloromethane	80 (A,W)	ID		<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
1,2-Dibromoethane	NC	NC		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
Chlorobenzene	100	25		<1	<1	<1	0.53	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
1,1,1,2-Tetrachloroethane	77	ID		<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1

Dataframe:

	NaN	NaN	Jate Collect	2d	02/25/2020	02/25/2020	02/25/2020	02/28/2020	02/25/2020	02/25/2020	...	02/27/2020	C
0	METALS ANALY iTS (ae)	DW	GSI	GSI	GSI	GSI	GSI	GSI	GSI	GSI	...	GSI	I
1	Arsenic, Dissolved	10	10	10	<0.002	<0.002	<0.002	| 4	<0.002	<0.002	...	— 2B	
2	Arsenic	10	10	10	<0.002	<0.002	<0.002	— 46	<0.002	0.316	...	62	
3	Chromium, Dissolved	100	160	G	0.275	0.891	7	— 0.204	0.562	0.463	...	13	
4	Chromium	100	160	G	0.42	2.153	6	~ 0.388	16	280	...	0.97	
5	Chromium VI, Dissolved	100	160	160	<0.005	<0.005	8	<0.01	<0.005	<0.005	...	<0.005	
6	Chromium VI	100	160	160	<0.01	<0.01	<0.01	/ <0.02	<0.01	<0.01	...	<0.5	
7	Copper, Dissolved	1000	20	G	0.405	<0.005	<0.005	— <0.005	3.473	0.396	...	0.751	
8	Copper	1000	20	G	<0.005	0.698	0.516	| 0.528	9	6	...	6	
9	Lead, Dissolved	4	44	G	<0.003	<0.003	<0.003	- 0.208	<0.003	<0.003	...	<0.003	
10	Lead	4	44	G	0.249	0.283	<0.003	» <0.003	<0.003	<0.003	...	4	
11	Selenium, Dissolved	50	5	5	<0.005	<0.005	<0.005	— <0.005	<0.005	9	...	<0.005	
12	Selenium	50	5	5	<0.005	<0.005	<0.005	— <0.005	<0.005	9	...	<0.005	
13	Zinc, Dissolved	2400	260	G	6	4.654	5	— 2.299	9	1.504	...	2	
14	Zinc	2400	260	G	1.56	2.074	1.632	2.05	14	1.948	...	44	
15	IVVOGCTANAIAY TE(ua/Ls)	DW	GSI	GSI	GSI	GSI	GSI	GSI	GSI	GSI	...	GSI	II
16	Diethyl ether	10 (E)	ID	ID	<10	<10	<10	| <10	<10	<10	...	<10	
17	Acetone	730	1,700	1,700	<50	<50	<50	| <50	<50	<50	...	<50	
18	Methyl iodide	NC	NC	NC	<4	<1	<1	<4	<4	<1	...	<1	
19	Carbon disulfide	800	NC	NC	<5	<5	<5	| 0.26	<5	<5	...	<5	
20	tert-Methyl butyl ether (MTBE)	AO (E)	7,100 (X)	7,100 (X)	<5	<5	<5	| <5	<5	<5	...	<5	
21	Acrylonitrile	2.6	2.0 (M); 1.2	2.0 (M); 1.2	<2	<2	<2	<2	<2	<2	...	<2	
22	2-Butanone (MEK)	13,000	2,200	2,200	<25	<25	<25	— <25	<25	<25	...	<25	
23	Dichlorodifluoromethane	1,700	ID	ID	<5	<5	<5	<5	<5	<5	...	<5	
24	Chloromethane	260	NC	NC	0.15	<5	<5	a.	<5	<5	...	<5	

Input image 2:

Table 1
Groundwater Analytical Results
RACER - Flint West # 12990

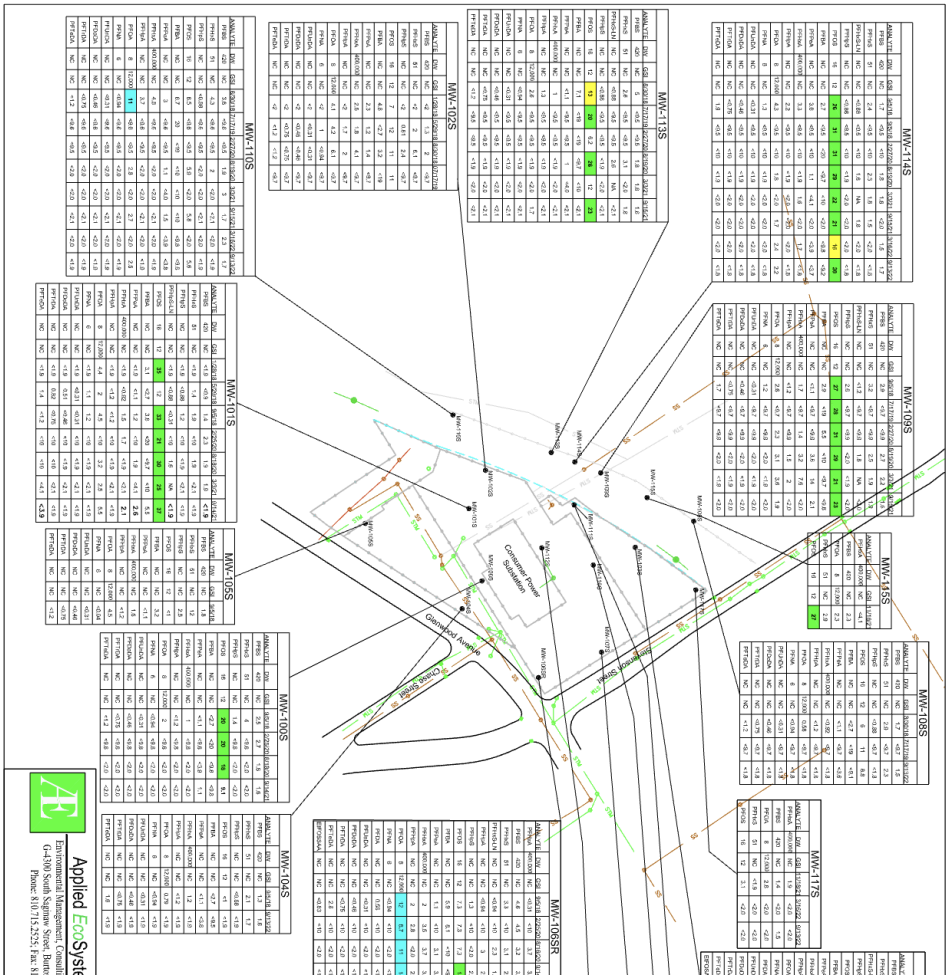
		Sample ID		MW-100S	MW-101S	MW-102S	MW-103S	MW-104S	MW-105S	MW-106SR	MW-107S	MW-108S	MW-109S	MW-110S	MW-111S	MW-112S	MW-113S	Field Dupe	Trip Blank	Field Blank	Equip Blank
		Date Collected		5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18	5/29/18		5/29/18	5/29/18
METALS ANALYTE (ug/L)	**DW**	GSI		DRY				DRY		DRY		NS	NS	NS			NS	MW-102S			
Arsenic (dissolved)	10	10					1.41									23					
Arsenic	10	10					9									60					
Chromium (dissolved)	100	160	G		1.62	14	0.168		1.01		0.328				4.13	0.431		14			
Chromium (total)	100	160	G		2.64	14	0.276		44		0.208				4.17	0.296		14			0.095
Chromium VI (dissolved)	100	160																			
Chromium VI (total)	100	160																			
Copper (dissolved)	1000	20	G		0.381	0.498			0.816		1.73				0.611			0.469			
Copper	1000	20	G		0.748	0.341			1.78		1.43				2.42						
Lead (dissolved)	4	44	G																		
Lead	4	44	G		0.408		0.068		0.304		0.121				0.067	0.062					
Selenium (dissolved)	50	5																			
Selenium	50	5																			
Zinc (dissolved)	2400	260	G		6	8	7		8		12				15	8		9			
Zinc	2400	260	G		6	7	8		7		8				6	4.76		9			1.79
VOC ANALYTE (ug/L)	**DW**	GSI																			
Acetone	730	1,700			5	4.5	6.4		5.6						5.9	8.2		5.6	4.8		8.4
Methyl iodide	NC	NC																			
Carbon disulfide	800	NC																			
2 Butanone (MEK)	13,000	2,200																			
Chloromethane	260	NC																			
Vinyl Chloride	2.0	13														9					
Chloroethane	430	1,100																			
trichlorofluoromethane	2,600	NA																			
1,1-Dichloroethene	7.0	130														1					
Methylene Chloride	5.0	1,500														0.17					
trans-1,2-Dichloroethene	100	1,500														0.25					
1,1-Dichloroethane	880	740														1					
cis-1,2-Dichloroethene	70	620														3					
Tetrahydrofuran	95	11,000																	4.8		
Chloroform	80	350																			0.46
1,1,1-Trichloroethane	200	89																			
4-Methyl-2-pentanone (MIBK)	1800	ID																			
2-Hexanone	1000	ID																			
Carbon tetrachloride	5.0	38 (X)																			
Benzene	5.0	200														0.20					
Bromodichloromethane	80.0	NC																			
Trichloroethene	5.0	200			1	0.67									7	13		0.7			
Toluene	790	270				0.39															
Tetrachloroethene	5.0	60							45												
Chlorobenzene	100	25																			
Styrene	100	80																			

1 of 4

Dataframe:

	NaN	NaN	Date Collect	ed	5/29/18	5/29/18	5/29/18	} 5/29/18	5/29/18	5/29/18	...	5/29/18	5/29/18	} 5/29/18	5/29/18
0	METALS*ANALYTE (ug/L)	DW	GSI	GSI	IDRY,	IDRY,	IDRY,	IDRY,	DRY,	DRY,	...	gS	INS)	INS)	INS)
1	— Argenic (dissolved) ts″	10	10	10	10	10	10	| 1.44	| 1.44	| 1.44	...	| 1.44	| 1.44	| 1.44	| 1.44
2	Arsenic	10	10	10	10	10	10	| 9g	| 9g	| 9g	...	| 9g	| 9g	| 9g	| 9g
3	Chromium (dissolved)	100	160	G	G	1.62	14	| 0.168	| 0.168	| 1.01	...	0.328	0.328	0.328	4.13
4	Chromium (total)	100	160	G	G	2.64	14	| 0.276	| 0.276	| 44	...	0.208	0.208	0.208	4.17
5	Chromium VI (dissolved)	100	160	160	160	160	160	160	160	160	...	160	160	160	160
6	Chromium VI (total)	100	160	160	160	160	160	160	160	160	...	160	160	160	160
7	Copper (dissolved)	1000	20	G	G	0.381	0.498	0.498	0.498	. 0.816	...	1.73	1.73	1.73	0.611
8	Copper	1000	20	G	G	0.748	0.341	0.341	0.341	| 1.78	...	1.43	1.43	1.43	2.42
9	Lead (dissolved)	4	44	G	G	G	G	G	G	G	...	G	G	G	G
10	Lead	4	44	G	G	0.408	0.408	| 0.068	| 0.068	| 0.304	...	0.121	0.121	0.121	0.067
11	Selenium (dissolved)	50	5	5	5	5	5	5	5	5	...	5	5	5	5
12	Selenium	50	5	5	5	5	5	5	5	5	...	5	5	5	5
13	Zinc (dissolved)	2400	260	G	G	6	8	| 7	| 7	| 8	...	12	12	12	15
14	Zinc	2400	260	G	G	6	7	| 8	| 8	| 7	...	8	8	8	6
15	VOCTANALY,TE (ug/L)	DW	GSI	GSI	GSI	GSI	GSI	GSI	GSI	GSI	...	GSI	GSI	GSI	GSI
16	Acetone	730	1,700	1,700	1,700	5	4.5	| 6.4	| 6.4	| 5.6	...	| 5.6	| 5.6	| 5.6	5.9
17	Methyl iodide	NC	NC	NC	NC	NC	NC	NC	NC	NC	...	NC	NC	NC	NC
18	Carbon disulfide	800	NC	NC	NC	NC	NC	NC	NC	NC	...	NC	NC	NC	NC
19	2 Butanone (MEK)	13,000	2,200	2,200	2,200	2,200	2,200	2,200	2,200	2,200	...	2,200	2,200	2,200	2,200
20	Chloromethane	260	NC	NC	NC	NC	NC	NC	NC	NC	...	NC	NC	NC	NC
21	Vinyl Chloride	2.0	13	13	13	13	13	13	13	13	...	13	13	13	13
22	Chloroethane	430	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	...	1,100	1,100	1,100	1,100
23	trichlorofluoromethane	2,600	2,600	2,600	2,600	2,600	2,600	2,600	2,600	2,600	...	2,600	2,600	2,600	2,600
24	1,1-Dichloroethene	7.0	130	130	130	130	130	130	130	130	...	130	130	130	130
25	Methylene Chloride	5.0	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	...	1,500	1,500	1,500	1,500
26	- trans-1,2-Dichloroethene	100	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	...	1,500	1,500	1,500	1,500
27	1,1-Dichloroethane	880	740	740	740	740	740	740	740	740	...	740	740	740	740

Input image 3:



My algorithm was able to detect 12 tables

```
### page_3_table_0
```

	Unnamed: 0	N	IW -	lOo8s	Unnamed: 4	Unnamed: 5
0	ANALYTE PFBS	_ DW_ 420	_GSI_ NG	8/30/18 AT	FA7I9 <97	9/13/22 15
1	PFHxS	51	NC	2.9	<9.7	2.3
2	PFHpS	NC	NC	<0.88	<9.7	<1.8
3	PFOS	16	12	6	11	8.8
4	PFBA	NC	NC	<2.7	<19	<9.1
5	PFPeA	NC	NC	<1.1	<9.7	<3.6
6	PFHxA	400,00(	NC	<0.92	<9.7	<1.8
7	PFHpA	NC	NC	<1.2	<9.7	<1.8
8	PFOA	8	12,000	0.58	<9.7	<1.8
9	PFNA	6	NC	<0.94	<9.7	<1.8
10	PFUnDA	NC	NC	<0.31	<9.7	<1.8
11	PFDoDA	NC	NC	<0.46	<9.7	<1.8
12	PFTrDA	NC	NC	<0.75	<9.7	<1.8
13	PFTeDA	NC	NC	<1.2	<9.7	<1.8

page_3_table_1

	Unnamed: 0	Unnamed: 1	Unnamed: 2	M\	V-10!	S	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	ANALYTE PFBS	DW 420	GSI NC	2/5/18 29	27/19 <O.7	2127120 <9.9	8/19/20 27	313/21. 22	9/15/24 16
1	PFHxS	51	NC	3.2	<9.7	<9.9	25	1.9	1.9
2	PFHxS-LN	NC	NC	<1.2	<9.7	<9.9	18	NaN	<2.0
3	PFHpS	NC	NC	2.6	<9.7	<9.9	<2.0	<1.9	<2.0
4	PFOS	16	12	27	28	31	29	21	23
5	PFBA	NC	NC	29	<19	5.5	<10	<9.7	<9.8
6	PFPeA	NC	NC	<1.	<9.7	<9.9	3.6	14	2.1
7	PFHxA	400,000	NC	1.7	<9.7	1.4	3.2	76	<2.0
8	PFHpA	NC	NC	<1.2	<9,7	<9.9	1.5	2	<2.0
9	PFOA	8	12,000	2.6	<9.7	2.3	3.1	3.6	1.9
10	PFNA	6	NC	1.2	<9.7	<9.9	<2.0	<1.9	<2.0
11	PFUnDA	NC	NC	<0.31	<9.7	<9.9	<2.0	<1.9	<2.0
12	PFDoDA	NC	NC	<0,46	<9,7	<9.9	<2.0	<1.9	<2.0
13	PFTrDA	NC	NC	<0.75	<9.7	<9.9	<2.0	<1.9	<2.0
14	PFTeDA	NC	NC	1.7	<9.7	<9.9	<2.0	<1.9	<2.0

page_3_table_2

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	M\	V-174.	iS	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: 10
0	ANALYTE PFBS:	_ DW 420	GSI NC	9/518 A?	SISHB <9.5	2/27/20 <10	8/19/20 18	RKAn 18	SA5/21 <2.0	3/16/22 15	9/13/22 V7
1	PFHxsS	51	NC	2.4	<9.5	<10	2.3	1.8	1.5	<2.0	<1.8
2	PFHxS-LN	NC	NC	<0.88	<9.5	<10	1.6	NaN	1.8	<2.0	<1.8
3	PFHpS	NC	NC	<0.88	<9.5	<10	<1.9	<2.0	<2.0	<2.0	<1.8
4	PFOS	16	12	26	31	31	29	22	21	16	30
5	PFBA	NC	NC	27	<9.5	<20	<9.7	<10	<2.0	<9.8	<9.2
6	PFPeA	NC	NC	3.6	<9.5	<10	14	<4.1	<2.0	<3.9	<3.7
7	PFHxA	400,00(	NC	3.3	<9.5	<10	<1.9	1.6	<2.0	1.7	<1.8
8	PFHpA	NC	NC	2.2	<9.5	<10	<1.9	<2.0	<2.0	<2.0	<1.8
9	PFOA	8	12,000	43	<9.5	<10	1.8	<2.0	1.7	2.4	2.2
10	PFNA	6	NC	1.3	<9.5	<10	<1,.9	<2.0	<2.0	<2.0	<1,.8
11	PFUnDA	NC	NC	<0.31	<9.5	<10	<1.9	<2.0	<2.0	<2.0	<1.8
12	PFDoDA	NC	NC	<0.46	<9.5	<10	<1.9	<2.0	<2.0	<2.0	<1.8
13	PFTrDA	NC	NC	<0.75	<9.5	<10	<1.9	<2.0	<2.0	<2.0	<1.8
14	PFTeDA	NC	NC	1.9	<9.5	<10	<1,.9	<2.0	<2.0	<2.0	<1,.8

page_3_table_3

	NV	\W-1	15S	Unnamed: 3
0	ANALYTE PFHxA	_DW_ 400,00¢	_GSL NC	14/1872" <41
1	PFBS	420	NC	2.3
2	PFOA	8	12,000	2.3
3	PFHxS	51	NC	2.9
4	PFOS	16	12	27

page_3_table_4

	Unnamed: 0	Unnamed: 1	/W-	117°	Unnamed: 4	Unnamed: 5
0	ANALYTE PFHxA	DW 400,00C	_GSL NG	11/18/21 19	3/16/22 <2.0	9/13/22, <2.0
1	PFBS	420	NC	1.4	<2.0	1.5
2	PFOA	8	12,000	2.8	<2.0	<2.0
3	PFHxS	51	NC	<1.9	<2.0	<2.0
4	=, PFOS	16	12	3.1	<2.0	<2.0

page_3_table_5

	Unnamed: 0	Unnamed: 1	Unnamed: 2	M\	V-11:	3S	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	ANALYTE PFBS	_Dw_ 420	_GSL NG	8/30/18 §	279 <9.5	2127120 <9.5	8/19/20 18	3ISI2L. 18	9/15/24 1.6
1	PFHxS	51	NC	2.6	<9.5	<9.5	3.1	<2.0	1.8
2	PFHxS-LN	NC	NC	<0.88	<9.5	<9.5	26	NaN	<2.1
3	PFHpS	NC	NC	<0.88	<9.5	<9.5	<1.9	<2.0	<2.1
4	PFOS	16	12	13	20	6.2	26	12	23
5	PFBA	NC	NC	7.1	<19	<19	<9.7	<10	<2.1
6	PFPeA	NC	NC	<1.1	<9.5	<9.5	1	<4.0	<2.1
7	PFHxA	400,00¢	NC	1	<9.5	<9.5	<1.9	<2.0	<2.1
8	PFHpA	NC	NC	1.3	<9.5	<9.5	<1.9	<2.0	<2.1
9	PFOA	8	12,000	2.6	<9.5	<9.5	<1.9	<2.0	1.7
10	PFNA	6	NC	<0.94	<9.5	<9.5	<1.9	<2.0	<2.1
11	PFUnDA	NC	NC	<0.31	<9.5	<9.5	<1.9	<2.0	<2.1
12	PFDoDA	NC	NC	<0.46	<9.5	<9.5	<1.9	<2.0	<2.1
13	PFTrDA	NC	NC	<0.75	<9.5	<9.5	<1.9	<2.0	<2.1
14	PFTeDA	NC	NC	<1.2	<9.5	<9.5	<1.9	<2.0	<2.1

page_3_table_6

	ANALYTE PFHxA	_DW_ 400,000	_GSL NC	S158 <0.31	2125120 <10	8/18/20 37
0	PFBS	420	NC	4.6	4.5	3.2
1	PFHxS	51	NC	3.3	<10	3.1
2	PFHxS-LN	NC	NC	<0.94	<10	2.3
3	PFHpA	NC	NC	<0.94	<10	3
4	PFHpS	NC	NC	1.3	<10	<2.0
5	PFOS	16	12	7.3	7.3	7.3
6	PFBA	NC	NC	5.9	8.1	<10
7	PFPeA	NC	NC	1.1	<10	3.1
8	PFHxA	400,00C	NC	2	3.5	3.7
9	PFHpA	NC	NC	2	28	<2.0
10	PFOA	8	12,000	12	8.7	11
11	PFNA	6	NC	<0.94	<10	<2.0
12	PFDA	NC	NC	0.58	<10	<2.0
13	PFUnDA	NC	NC	<0.31	<10	<2.0
14	PFDoDA	NC	NC	<0.46	<10	<2.0
15	PFTrDA	NC	NC	<0.75	<10	<2.0
16	PFTeDA	NC	NC	2.8	<10	<2.0
17	EtFOSSAA	NC	NC	<0.83	<10	<2.0

page_3_table_7

	ANALYTE PFBS	_DWw_ 420	GSI NG	128/18 <2	5/29/18 13	8/30/18 2	PATHE <O7
0	PFHxS	51	NC	<2	2	6.1	<9,7
1	PFHpS	NC	NC	<2	0.81	2.4	<9.7
2	PFOS	16	12	7	12	11	11
3	PFBA	NC	NC	48	<2.7	3.2	<19
4	PFPeA	NC	NC	2.3	1.2	1.4	<9,7
5	PFHxA	400,00¢	NC	25	1.8	41	<9.7
6	PFHpA	NC	NC	<2	1.7	2	<9.7
7	PFOA	8	12,000	4.1	4.2	6.1	<9.7
8	PFNA	6	NC	<2	1	<0.94	<9,7
9	PFUnDA	NC	NC	<2	<0.31	<0.31	<9.7
10	PFDoDA	NC	NC	<2	<0.46	<0.46	<9.7
11	PFTrDA	NC	NC	<2	<0.75	<0.75	<9.7
12	PFTeDA	NC	NC	<2	<1.2	<1.2	<9,7

page_3_table_8

	Unnamed: 0	MMU	-10.	RS	Unnamed: 4
0	ANALYTE PFBS	_DW_ 420	_GSL NC	91518. 1.3	9113/22 16
1	PFHxS	51	NC	2.1	1.7
2	PFHpS	NC	NC	<0.88	<1.9
3	PFOS	16	12	<1	<1.9
4	PFBA	NC	NC	<2.7	<9.5
5	PFPeA	NC	NC	<1.1	<3.8
6	PFHXxXA	400,00¢	NC	1.2	<1.9
7	PFHpA	NC	NC	<1.2	<1.9
8	PFOA	8	12,000	0.79	<1.9
9	PFNA	6	NC	<0.94	<1.9
10	PFUnDA	NC	NC	<0.31	<1.9
11	PFDoDA	NC	NC	<0.46	<1.9
12	PFTrDA	NC	NC	<0.75	<1.9
13	PFTeDA	NC	NC	1.6	<1.9

page_3_table_9

	Unnamed: 0	Unnamed: 1	MV	/-10(	IS	Unnamed: 5	Unnamed: 6
0	ANALYTE PFBS	DW 420	_GSL NC	1518. 25	2/25/20 27	8/18/20 16	9/14/24 16
1	PFHxS	51	NC	4	<9.8	<2.0	<2.0
2	PFHpS	NC	NC	1.4	<9.8	<2.0	<2.0
3	PFOS	16	12	20	20	18	9.1
4	PFBA	NC	NC	<2.7	<20	<9.8	<9.8
5	PFPeA	NC	NC	<1.1	<9.8	<3.9	1.1
6	PFHxA	400,000	NC	1	<9.8	<2.0	<2.0
7	PFHpA	NC	NC	<1.2	<9.8	<2.0	<2.0
8	PFOA	8	12,000	2	<9.8	<2.0	<2.0
9	PFNA	6	NC	<0,94	<9.8	<2.0	<2.0
10	PFUnDA	NC	NC	<0.31	<9.8	<2.0	<2.0
11	PFDoDA	NC	NC	<0.46	<9.8	<2.0	<2.0
12	PFTrDA	NC	NC	<0.75	<9.8	<2.0	<2.0
13	PFTeDA	NC	NC	<1.2	<9.8	<2.0	<2.0

page_3_table_10

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Mi\	V-111	S	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: 10
0	ANALYTE PFBS	_DW_ 420	_GSL NC	8/30/18 3.6	FA7AG <9.6	2/27/20 <O.5	8/19/20 19	3/3/21. 3	9/15/21 AT	3/16/22 23	9/13/22 AT
1	PFHxS	51	NC	4.3	<9.6	<9.5	2	<2.0	<2.1	<2.0	<1.9
2	PFHpS	NC	NC	<0.88	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
3	PFOS	16	12	8.5	<9.6	<9.5	5.9	<2.0	5.8	<2.0	5.6
4	PFBA	NC	NC	6.7	20	<19	<10	<10	<10	<9.8	<9.6
5	PFPeA	NC	NC	3	<9.6	<9.5	1.1	<4.0	15	<3.9	<3.8
6	PFHXA	400,000	NC	48	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
7	PFHpA	NC	NC	3.7	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
8	PFOA	8	12,000	1	<9.6	<9.5	2.8	<2.0	27	<2.0	25
9	PFNA	6	NC	<0.94	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
10	PFUnDA	NC	NC	<0.31	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
11	PFDoDA	NC	NC	<0.46	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
12	PFTrDA	NC	NC	<0.75	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9
13	PFTeDA	NC	NC	<1.2	<9.6	<9.5	<2.0	<2.0	<2.1	<2.0	<1.9

page_3_table_11

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	AW -'	01S	Unnamed: 6	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: 10	\¥	VAS24	NAS	Unnamed: 14
0	ANALYTE, PFBS	_DW_ 420	_GSI_ NG	1/28/18 <19	5/29/18 <0.9	9/5/18 _ 14	2/25/20 23	8/18/20 19	3/3/21 19	9/14/21 <19	NaN	ANALYTE PFBS:	_DW_ 420	_GSL NC	9/518. 1.8
1	PFHxS	51	NC	<1.9	1.4	1.9	<10	1.9	<2.1	<1.9	NaN	PFHxS	51	NC	12
2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	PFHpS	NC	NC	<1,9	<0.88	1.2	<10	<1,9	<2.1	<1,9	NaN	PFHpS	NC	NC	25
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
5	PFHpS-LN	NC	NC	<1.9	<0.88	<0.31	<10	1.6	NaN	<1.9	NaN	PFOS	16	12	<1
6	PFOS	16	12	35	12	33	21	30	25	37	NaN	PFBA	NC	NC	32
7	PFBA	NC	NC	3.1	<2.7	3.8	<20	<9.7	<10	5.5	NaN	PFPeA	NC	NC	<4.1
8	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
9	PFPeA	NC	NC	<1,9	<1	1.2	<10	1.9	<41	2.6	NaN	PFHxA	400,00C	NC	1.6
10	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
11	PFHxA	400,000	NC	<1.9	<0.92	1.5	1.7	<1.9	<2.1	2.1	NaN	PFHpA	NC	NC	<1.2
12	PFHpA	NC	NC	<1.9	<1.2	<1.2	<10	<1.9	<2.1	<1.9	NaN	PFOA	8	12,000	4.5
13	PFOA	8	12,000	44	2	45	<10	3.2	25	5.5	NaN	PFNA	6	NC	<0.94
14	PFNA	6	NC	<1.9	11	1.2	<10	<1.9	<2.1	<1.9	NaN	PFUnDA	NC	NC	<0.31
15	PFUnDA	NC	NC	<1.9	<0.31	<0.31	<10	<1.9	<2.1	<1.9	NaN	PFDoDA	NC	NC	<0.46
16	PFDoDA	NC	NC	<1.9	0.51	<0.46	<10	<1.9	<2.1	<1.9	NaN	PFTrDA	Nc	Nc	<0.75
17	PFTrDA	NC	NC	<1.9	0.82	<0.75	<10	<10	<2.1	<1.9	NaN	PFTeDA	NC	NC	<4.2
18	PFTeDA	NC	NC	<1.9	1.4	<1.2	<10	<10	<4.1	<3.9	NaN	NaN	NaN	NaN	NaN

Steps to run:

python -m venv venv
source venv/bin/activate

pip install -r requirements.txt

macOS
brew install tesseract

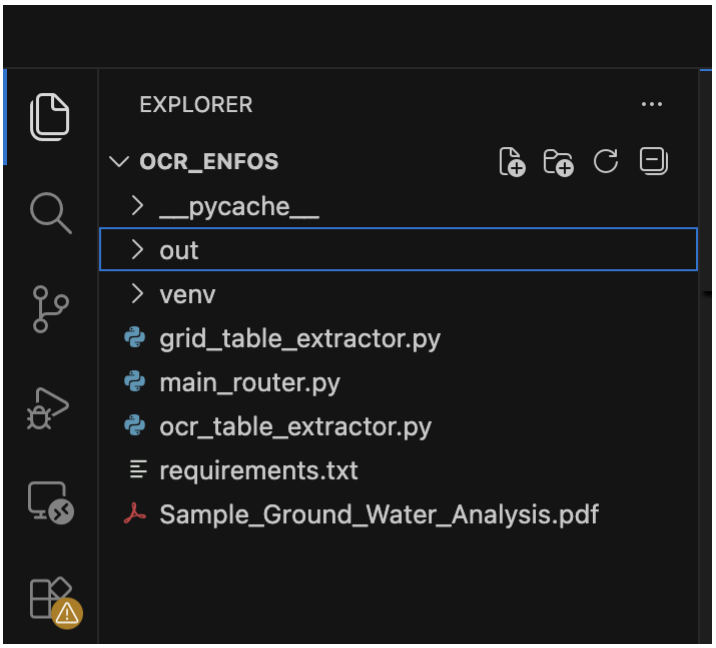
# Ubuntu
sudo apt install tesseract-ocr

Windows
Download from: https://github.com/tesseract-ocr/tesseract

There are many places where you can download the latest version of Tesseract OCR. One such place is from https://github.com/UB-Mannheim/tesseract/wiki. Download the 64 bit version.

command: python3 main_router.py Sample_Ground_Water_Analysis.pdf

Folder structure:



Intelligent Page Routing Layer for Adaptive Table Extraction

main_router.py:

The file implements a high-performance orchestration layer that dynamically routes each page of a document to the most appropriate extraction strategy based on its structural characteristics. Rather than applying a single monolithic pipeline, this module enables adaptive, per-page decision-making, significantly improving robustness across heterogeneous document types.

System overview

At a high level, this component functions as a **lightweight document understanding router**, sitting between raw PDF ingestion and specialized extraction engines:

1. **Input:** Multi-page PDF document
2. **Processing:** Page-wise structural classification
3. **Output:** Structured tables (CSV + LLM-ready Markdown + metadata)

It integrates two fundamentally different extraction paradigms:

1. **Grid-aware extraction** (for ruled/tabular layouts)
2. **OCR-driven reconstruction** (for weakly structured or scanned layouts)

This separation of concerns allows the system to **optimize for both precision (structured tables)** and **recall (unstructured OCR-heavy pages)**.

Core Innovation: Structural Signal-Based Page Classification

The system uses line density as a lightweight structural heuristic to determine how each page should be processed.

Each page is:

1. Converted to grayscale and binarized using Otsu thresholding
2. Processed with directional morphological filters to extract horizontal and vertical line structures

A normalized line density score is computed:

line_ratio = (horizontal_pixels + vertical_pixels) / total_pixels

This metric acts as a proxy for layout structure:

High line density - grid-based tables (structured extraction)
Low line density - weak/borderless layouts (OCR-based reconstruction)

This enables fast, interpretable, and model-free routing between extraction strategies, making the pipeline robust across diverse document formats.

grid_table_extractor.py:

This file implements a general-purpose table extraction pipeline that works across different types of documents, including both structured (grid-based) and unstructured (borderless) tables.

It uses a **hybrid computer vision + OCR pipeline** to extract tables from documents by leveraging low-level image features.

Core Processing Steps:

1. Preprocessing (Signal Enhancement)
    a. Convert image to **grayscale**
    b. Apply **Gaussian blur** to reduce noise
    c. Use **adaptive thresholding (binary inversion)** to highlight text and lines
2. Line Detection using Morphology
    a. Apply **horizontal and vertical kernels** using OpenCV morphological operations.
    b. Extract:
        i. horizontal lines - potential row boundaries
        ii. vertical lines - potential column boundaries

    These kernels are **scale-aware**, adapting to image size.
3. Table Region Detection
    a. Combine detected lines into a single mask
    b. Use **dilation + closing** to connect broken structures
    c. Detect contours to identify **table bounding boxes**
    d. Merge overlapping regions using **IoU + proximity**
4. Adaptive Extraction Strategy
    a. Based on line density:
        i. High line density → Grid-based extraction
            1. Detect row/column separators from line projections
            2. Form a grid and extract each cell
            3. Apply cell-level OCR
        ii. Low line density → OCR-based reconstruction
            1. Use **Tesseract word bounding boxes**
            2. Cluster words by:
                vertical alignment → rows
                horizontal alignment → columns
            3. Reconstruct table structure without explicit borders
5. OCR and Cleanup
    a. Apply **adaptive thresholding again at cell level** for better OCR accuracy
    b. Use **Tesseract (LSTM engine)** to extract text
    c. Clean text (whitespace normalization, noise removal)
    d. Filter out sparse/empty columns

Output

1. Structured CSV files per table
2. Unified LLM-ready Markdown file
3. JSON summary for traceability

<u>ocr_table_extractor.py</u> (Fallback Engine)

This module handles **complex or messy pages** where clear table grids are not present. It is designed to extract tables from **scanned, rotated, or borderless documents** using OCR and spatial reasoning.

1. Preprocessing (rotation + deskew)
    a. detects page orientation using Tesseract (OSD)
    b. corrects rotation (90°, 180°, etc.)
    c. applies **fine deskewing** using OpenCV to align text horizontally
2. Table Detection using Bounding Boxes
    a. Uses img2table to detect table regions
    b. Each table is represented as a **bounding box (x1, y1, x2, y2)**
    c. These coordinates are used to **crop and isolate table regions**
3. Recursive Extraction Strategy
    a. For each detected region:
        i. first, attempt **grid-based extraction** (from the main pipeline)
        ii. If that fails, fall back to **OCR-based reconstruction**

    This ensures we always use the **most reliable method available**.

4. OCR + Spatial Reconstruction
    a. OCR detects word-level bounding boxes
    b. Each word has coordinates defining its position

        Using these coordinates:

        1. Words aligned along similar y-values → grouped into rows
        2. Words aligned along similar x-values → grouped into columns

Therefore, table structure is inferred from **spatial alignment, not visual borders**

5. Post processing
    a. Clean text and remove noise
    b. Drop empty rows/columns
    c. Ensure column names are valid and unique

Output

1. CSV files for each table
2. Cropped table images (for debugging)
3. Structured summary with metadata

Final Ouput

llm_tables.md
which is given as input to an LLM

Future Work

If given the position, the first thing I would do is audit every failure case in the current pipeline. I would check every page where the router fell back to full-page OCR, every CSV that came out misaligned, every borderless table that img2table missed and use those failures to build a labeled evaluation set. That set becomes the ground truth that all future work is measured against. From there, I would replace the fragmented stack of Tesseract, img2table, and TATR with a single fine-tuned vision transformer trained specifically on environmental report layouts, using synthetic table generation to bootstrap training data before touching any real documents. The goal in the first few days would not be to build something new, it would be to make the existing pipeline reliable on the document types the company actually processes.

Once the pipeline is stable, I would shift focus to the embedding layer. Right now the output is CSVs which are basically flat files that discard spatial and semantic relationships the extractor had to work hard to recover. I would replace that with a cell-level vector store where each extracted value carries its row header, column header, table provenance, and a confidence score as part of its representation. That turns the extraction output into something a RAG system can actually reason over. Instead of chunking CSV rows into text and hoping the retriever finds the right one, you get structured retrieval where "give me all TCE readings above the DW standard for MW-107S" is a real query with a real answer.

The longer-term work would be pushing toward layout-agnostic extraction that would involve handling the documents that look nothing like conventional tables but are still making structured claims: site maps with embedded data grids, split-column forms, margin annotations, tables that continue across pages with no visual indication. That requires moving away from table detection as a geometry problem and toward something closer to semantic parsing. The end goal is a grid detector that is so thoroughly trained on every possible visual manifestation of structured data — bordered, borderless, rotated, degraded, embedded in maps, split across pages — that the concept of a 'failure case' effectively ceases to exist. A general enough grid detector is the solution.